

Ritabrata Patra

Kolkata, West Bengal

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Summary

Aspiring Computer Science student with a strong foundation in Java. Eager to apply academic knowledge and hands-on project experience in a real-world setting. Seeking a full-time role to contribute to impactful software solutions. Passionate about solving complex problems through code and open-source collaboration.

Skills

- **Languages:** Java, Python, SQL
- **AI / Machine Learning:** Scikit-learn (Random Forest, Logistic Regression), OpenCV, NLP basics, Model Evaluation
- **Full Stack Development:** HTML5, CSS3, JavaScript, React.js, Flask
- **Developer Tools:** Git, GitHub, VS Code, Jupyter Notebook
- **Core Competencies:** Data Structures (DSA), Problem Solving, Product Thinking, Research & Analysis, Team Leadership, Communication

Projects

Malware Detection using Machine Learning (Group Project – Major Contributor)

- Built a cybersecurity tool to detect malware in PE/.bat files and classify URLs as safe or malicious.
- Implemented Random Forest (95% accuracy) for PE analysis, heuristic rules for batch files, and Logistic Regression with NLP (88% accuracy) for URL classification.
- Designed a Tkinter GUI for real-time results, demonstrating applied AI in a practical security context.

Digital Document Signing Application (Independent Project)

- Developed a zero-dependency Java desktop application for digitally signing and managing documents.
- Built a freehand signature canvas, PDF import with metadata extraction (no external libraries), signer detail capture, and signed HTML export as a single-file Java Swing app.

Blockchain-based Face Recognition Voting System (Group Project – Major Contributor)

- Built a secure e-voting web application integrating facial recognition (OpenCV + KNN classifier) for voter authentication.
- Designed a custom proof-of-work blockchain to ensure tamper-proof, double-vote-free vote recording with live results.

Certifications

Introduction to Machine Learning

IIT Madras (NPTEL/Swayam)

- 12-week course · Jul–Oct 2025

Marketing Analytics – Elite

IIT Kharagpur (NPTEL/Swayam)

- 12-week course · Jan–Apr 2026

The Bits and Bytes of Computer Networking

Google, via Coursera

- Completed Nov 2025

Education

VIT Bhopal University

2023 – 2027

Bachelor's Degree, Computer Science — GPA: 7.91

National Gems School

2023

Class 12th, ICSE Board — 81%

National Gems School

2021

Class 10th, ICSE Board — 93.6%

Experience

Prodigal AI — Website Audit & Optimization Intern

Conducted website audits covering SEO, content, accessibility, and performance. Performed web scraping and brand research to extract and analyze website data.